INCH-POUND
MIL-PRF-49450/14(CR)
10 March 2000

PERFORMANCE SPECIFICATION SHEET

BATTERY CELL, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-648A/A

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49450(CR)

Performance.

Voltage. The voltage of the cell shall be 1.2 Vdc. (nominal)

Capacity. The capacity of the cell shall be a minimum of 18 ampere hours.

Weight. The weight of the cell shall be no more than 2.1 pounds.

Venting. The vent caps of completed cells shall vent gases, in such a manner, that the differential between the internal cell pressure and the external pressure of the surroundings shall not exceed 10 psig, but shall not vent gasses from the cell unless the differential between the internal and external pressure exceeds 2 psig.

AMSC N/A: FSC 6140

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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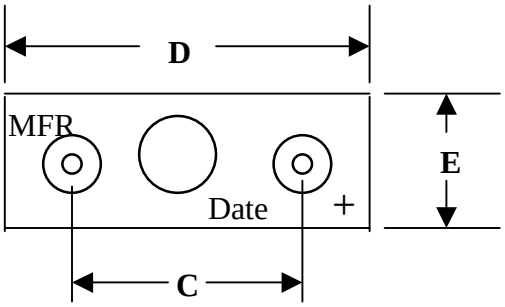
“A”, Dimension	4.890± .040
“B”, Dimension	5.425± .040
“C”, Dimension	1.878± .015
“D”, Dimension	3.110± .025
“E”, Dimension	1.645± .030



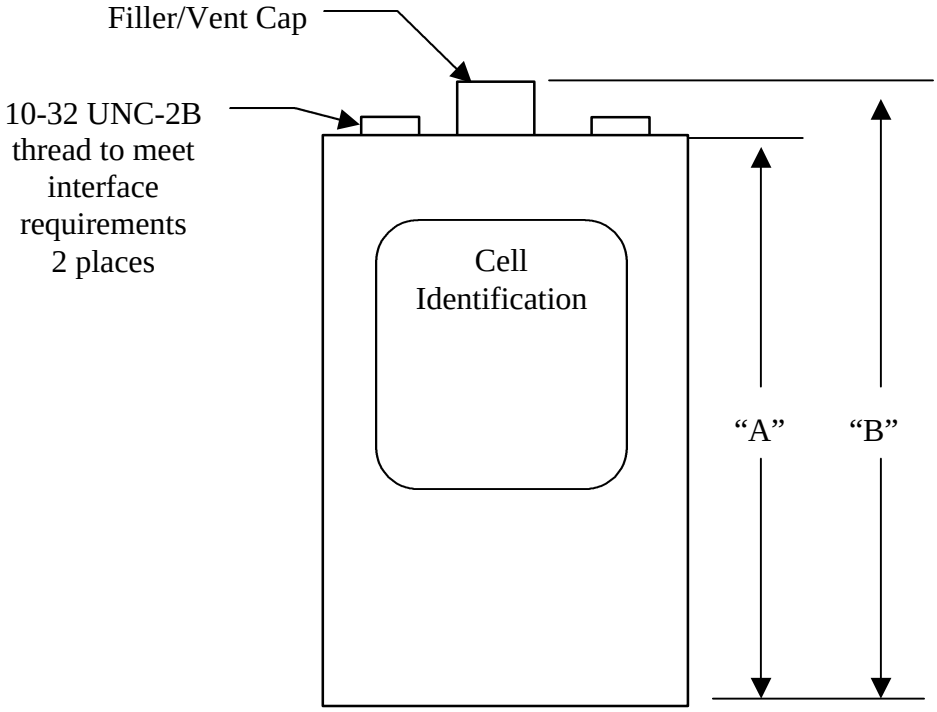


Figure 1. Battery Cell, BB-648A/A

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DRAWING NOTES:

All dimensions are in inches, unless specified otherwise.

Hardware. Each cell terminal shall be furnished with a set of hardware, tightened in accordance with applicable torque requirement. The terminal hardware shall consist of a hex head machine screw, a belleville spring and a washer. The hardware shall be resistant to 1.3 KOH. All components, as mentioned above, shall be compatible with each other.

Dimensions. The dimensional inspection of the cells shall be made after addition of electrolyte and meet the requirements of Fig. 1

Tongue and groove. If polyamide is utilized for cell containers, then the cell closure of cover to cell case shall be tongue and groove type.

Interchangeability. Relatively broad tolerances have been allowed for the cell. Nineteen (19) cells, taken from a production lot at random, must be able to be inserted in and removed from a battery case meeting the latest Army configuration without using a mallet or similar tool or other than a standard cell pulling device. Also sustained effort shall be made to minimize dimensional variations of the cells between production lots, and contracts.

Terminal seals. Completed cells shall withstand an internal pressure of 20 ±1 psig for 15 seconds when pressurized through the vent plug opening without cracking or rupturing in any manner.

Terminal Marking. Positive and negative terminal marking shall be embossed or impressed with "+" and "—", respectively.

Red Band The positive half of the cell cover shall be permanently dyed red. Minimum width of red band shall be 0.500 inches.

Shipping. All cells shall be shipped in a completely discharged condition with a shorting wire wrapped around both screws and the warning tag attached.

REQUIREMENTS:

Nomenclature:	BB-648A/A Battery Cell, Rechargeable
Rated capacity:	18 ampere-hours
Voltage (Nominal):	1.2 Volts
Weight (Maximum):	2.1 pounds

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Test	Discharge Rate(Amps)	Cut Off Voltage(V)	Capacity(AH) Minimum
Capacity	12	.95	18
Cycle Life			
Normal Rate	18	.95	16.2
High Rate	180	.76	12
Vibration	18	.95	18
Temperature Shock	18	.95	18
Altitude	18	.95	15.3
Humidity and Charge Retention	18	.95	13.5
High Rate			
Discharge and Low Temperature	180	.68	9
High Rate			
Discharge and Operating Position	180	.76	15.3
Shock	18	.95	18

Custodian:
Army - CR

Preparing activity
Army - CR

(Project 6140-A917-06)

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